

$V=15$ volts, the maximum achievable timing is 3686 seconds, which is slightly more than one hour. Frequencies as low as 0.1 Hz can be easily generated using the 555 timer. The timer can also withstand fluctuations in the supply voltage.

Here in the circuit, at 12V, resis-

tors R24, R25 and capacitor C2 produce around 1.47Hz frequency and the relay de-energises after around 1.5 hours to switch off the load. At the same time, the alarm built around IC4 sounds indicating that the appliance is off. The circuit can be assembled on any general-purpose PCB.

EFY note. Add two resistors to the potentiometer—one at the ground end and the other at the positive supply end—for minimum and maximum timing. Vary VR1 between 0 and V_{Omax} such that the op-amp output doesn't exceed the supply voltage ($V_{Omax} = V-1.5$). ●

MOTION SENSOR FOR SECURITY LIGHT

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Here is a system based on PIR motion detector module BS1600 (or BS1700) that can be used for security or corridor light-

The working of the circuit is simple. When you power-on the circuit after assembling all the components including the CFL, the CFL will glow for 10 seconds, turn off for 30 seconds, glow for 10 seconds and then turn off.

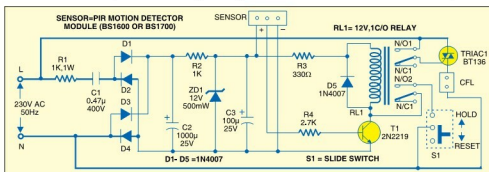


Fig. 1: Circuit of motion sensor for security light



Fig. 2: PIR motion detector module (BS1600 or BS1700)

ing in power-saving mode. The 12V DC power supply required for the motion detector and the relay driver is derived from 230V, 50Hz mains using a transformerless circuit as shown in Fig. 1.

Now the circuit is ready to work.

When any movement is detected, around 3.3V appears on the base of relay-driver transistor T1 and it conducts to energise relay RL1. As a result, TRIAC1 (BT136) fires to provide full 230V and light up the CFL. Another normally-opened contact of the relay (N/O2) is used here to hold the output until reset. If the switch is not in 'hold' position, the light will remain

'on' for about ten seconds (as programmed in the motion sensor). In short, when there is a movement near the sensor, the CFL glows for about ten seconds. It will remain 'on' if switch S1 is in 'hold' position.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Use a three-pin connector for connecting the PIR sensor in the circuit with correct polarity. The motion detector is embedded onto the transparent cover of the light assembly as shown in Fig. 2

An arrangement of CFL assembly in the author's prototype (Fig. 3) is shown in Fig. 4. In this arrangement, a PIR sensor and 23W, 230V AC CFL are used. Seal all four sides with Blue Tac for water-tightness. Insulate the track side of the PCB using an insulating foam and glue to the base. ●



Fig. 3: Author's prototype



Fig. 4: CFL assembly